

Youth Mental Wellness Platform using Fine-Tuned BERT for Suicidal Ideation Detection

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Abstract—The rising prevalence of mental health disorders among adolescents, coupled with barriers such as stigma, privacy concerns, and lack of accessibility, necessitates scalable digital interventions. This paper presents the design and implementation of a privacy-first mental wellness platform powered by Natural Language Processing (NLP). The core novelty of the system lies in its use of a fine-tuned BERT (Bidirectional Encoder Representations from Transformers) model to detect high-risk language and suicidal ideation with high recall. Unlike generic chatbots, this system integrates an empathetic AI companion, explainable mood analytics via topic modeling, and a tiered crisis escalation workflow.

We evaluate the model against robust traditional machine learning baselines (LinearSVC, Logistic Regression, Naive Bayes, Random Forest), demonstrating that our fine-tuned BERT model achieves a superior F1-score of 0.98, outperforming even the strongest baseline (LinearSVC, F1=0.94). The system architecture prioritizes user privacy through on-device inference capabilities and strict data minimization protocols, offering a secure environment for youth to seek support.

Index Terms—Mental Health, NLP, BERT, Suicide Prevention, Privacy-Preserving AI, Deep Learning

I. INTRODUCTION

The global landscape of adolescent health is currently witnessing a disturbing rise in mental health challenges. According to the World Health Organization, anxiety, depression, and feelings of isolation are reaching epidemic proportions among the youth demographic. Despite the critical nature of this public health crisis, a significant majority of young people actively avoid seeking professional assistance. This hesitation is frequently driven by deep-seated social stigma, profound concerns regarding personal privacy, and the logistical or financial barriers that make traditional therapy inaccessible.

To bridge this widening gap, we propose a sophisticated, privacy-centric mental wellness platform driven by Artificial Intelligence. While mobile and web-based digital interventions offer a promising avenue for providing low-friction support, the current market is saturated with fragmented applications that are often data-hungry, lack rigorous safety protocols, or rely on simplistic rule-based chatbots that fail to understand the nuance of human distress.

Consequently, this project introduces a holistic ecosystem that seamlessly integrates an empathetic AI companion, AI-enhanced journaling tools, and comprehensive mood analytics.

The primary contribution of this work is the implementation of a robust NLP pipeline utilizing fine-tuned BERT-family transformer classifiers. This model is specifically trained to identify suicidal ideation and high-risk language within unstructured text, enabling a "human-in-the-loop" crisis escalation mechanism that prioritizes user safety above all else.

II. RELATED WORK

The research foundation for this project draws upon a comparative analysis of existing digital mental health tools. Early innovations such as Woebot utilized Cognitive Behavioral Therapy (CBT) principles in a chatbot format. While successful in demonstrating that conversational agents could reduce symptoms of depression, its utility was largely restricted to scripted interactions without broader ecosystem features.

Subsequent platforms like Wysa improved accessibility through guided AI support but often relied heavily on cloud-based processing, which introduced potential privacy vulnerabilities. Similarly, peer forums on platforms like Reddit provide anonymity that reduces stigma but suffer from misinformation risks due to a lack of specialized moderation.

Our work builds upon the Transformer architecture introduced by Vaswani et al. and the BERT pre-training objectives described by Devlin et al. Unlike previous approaches that rely on keyword matching or shallow machine learning techniques, we leverage the contextual understanding of Transformers to distinguish between casual usage of negative words (e.g., "I'm dead tired") and genuine risk expressions (e.g., "I want to be dead").

III. METHODOLOGY AND SYSTEM DESIGN

A. System Overview

The platform is engineered as a Progressive Web App (PWA) that prioritizes a privacy-first architecture. The core system is a sophisticated NLP pipeline managing the flow from user input to actionable insight. Data ingestion from chat logs and journal entries is immediately sanitized and anonymized. This raw text is then subjected to preprocessing and tokenization before being fed into the Transformer models.

B. Theoretical Framework: The Transformer Architecture

To understand the superiority of our approach, it is essential to deconstruct the underlying Transformer architecture. Unlike Recurrent Neural Networks (RNNs) or LSTMs, which process data sequentially, Transformers process the entire sequence of text simultaneously using the Self-Attention mechanism.

The Self-Attention mechanism allows the model to weigh the importance of different words in a sentence relative to one another. Mathematically, the input vector is transformed into three vectors: Query (Q), Key (K), and Value (V). The attention score is calculated as:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (1)$$

Where d_k is the dimension of the key vectors. This scaling factor $\sqrt{d_k}$ prevents the dot products from growing too large, ensuring stable gradients during training. This mechanism allows our model to capture long-range dependencies, such as negations or context shifts that appear later in a sentence, which is critical for accurate risk detection.

C. Model Fine-Tuning

We utilized the `bert-base-uncased` model. The fine-tuning process involved adding a classification head consisting of a dropout layer (for regularization) and a linear layer mapping the hidden size (768) to the number of classes (2: Suicidal vs. Non-Suicidal).

The model was trained using Cross-Entropy Loss. To address class imbalance inherent in mental health datasets, we employed a class-weighted loss function, penalizing the model more heavily for missing positive cases (False Negatives).

D. Crisis Escalation Workflow

A defining feature of this platform is its tiered Crisis Escalation Workflow.

- **Tier 1 (Immediate Support):** If the Risk Confidence Score exceeds 0.85, the system suppresses standard generative responses and delivers a clinically validated empathetic message.
- **Tier 2 (Resource Connection):** The interface dynamically presents local emergency helplines and crisis resources.
- **Tier 3 (Human-in-the-Loop):** If the user consents, the interaction is flagged for human moderator review.

IV. EXPERIMENTS AND RESULTS

A. Dataset and Setup

We utilized the Kaggle Suicide and Depression Detection dataset, comprising approximately 232,000 samples scraped from moderated Reddit forums (r/SuicideWatch, r/depression). The dataset was split into 80% training and 20% testing sets. Text was preprocessed to remove URLs, user handles, and special characters, followed by tokenization using the BERT tokenizer.

B. Baseline Comparisons

To scientifically validate the Transformer-based approach, we implemented a comparative analysis against traditional Machine Learning algorithms. We selected four baselines utilizing TF-IDF (Term Frequency-Inverse Document Frequency) vectorization with a feature limit of 50,000 and class balancing enabled:

- 1) **Multinomial Naive Bayes (MNB):** A probabilistic classifier based on Bayes' theorem.
- 2) **Random Forest (RF):** An ensemble learning method using 200 estimators.
- 3) **Logistic Regression (LR):** A statistical model for binary classification.
- 4) **Linear SVC:** A Support Vector Machine optimized for linear classification.

C. Results Analysis

The models were evaluated based on Precision, Recall, F1-Score, and Accuracy. In the context of suicide prevention, **Recall** is the most critical metric, as missing a high-risk case (False Negative) can have fatal consequences.

Table I presents the comparative results using the test split.

TABLE I
PERFORMANCE COMPARISON OF ALGORITHMS

Model	Precision	Recall	F1-Score	Accuracy
Random Forest	0.90	0.91	0.90	89.98%
Naive Bayes	0.90	0.94	0.92	91.90%
Logistic Regression	0.95	0.93	0.94	94.22%
Linear SVC	0.95	0.94	0.94	94.37%
Fine-Tuned BERT	0.98	0.98	0.98	97.93%

As illustrated in Table I, the baselines (specifically Linear SVC and LR) performed strongly with F1-scores around 0.94. However, the Fine-Tuned BERT model outperforms the strongest baseline by approximately 3.5% in accuracy.

Analysis of Performance Gap: While the numerical gap appears small, it is statistically significant in error reduction. The BERT model reduces the misclassification rate from $\approx 5.6\%$ (Linear SVC) to $\approx 2.1\%$. Traditional models struggled with complex sentence structures involving negation or sarcasm (e.g., "I'm not going to kill myself"), whereas BERT's self-attention mechanism correctly interpreted these dependencies, reducing False Positives.

V. ETHICAL CONSIDERATIONS

Developing an AI for mental health requires adhering to the "Do No Harm" principle. Our system design implements a rigorous ethical safety net. We utilize Explainable AI (topic modeling) to ensure users understand why a mood trend is reported. Furthermore, in compliance with privacy norms, sensitive inference is performed on sanitized text, and no raw journals are stored without explicit, opt-in consent.

VI. CONCLUSION

This project demonstrates the viability of using Transformer-based deep learning for critical mental health monitoring. The superior performance of BERT over robust ML baselines validates the necessity of contextual embeddings in risk detection. By integrating this high-recall model into a privacy-first, human-in-the-loop application, we provide a scalable solution to the youth mental health crisis. Future work will focus on federated learning to further enhance privacy and the integration of multimodal signals such as voice intonation.

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